Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “motor” and “generator” to explain your answer. Sentence starters are optional.

motor: a device converting electrical energy to mechanical output

generator: a device converting mechanical input to electrical output

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A bar magnet is cut into **3** pieces. Does this produce isolated *north* and *south* poles? **Sketch the usual result**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives **20 J** electrically and delivers **12 J** mechanically; **8 J** goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp. **Label each input** and useful output.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A hand-cranked generator receives **50 W** of mechanical power and supplies **12 V** at **3 A**. **Complete the energy-rate account**. **Find electrical output power**, efficiency and the rate of transfer to other forms.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A motor drives a generator that feeds the motor. **Explain why real losses prevent** the loop from running forever without external input.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Move from r to **4** r away from a point charge. **Compare field magnitude** and say what you treated as true about the source charge.
Label the source charge, distances and electric-field directions.

Word help: charge = a property responsible for electric interactions

Sentence starter: Both ... . They differ in ... because ... .



Source charge Field location

Marzano check - circle: 1 2 3

07 Exit check

A motor is **80%** efficient and a generator is **75%** efficient. **Find their combined useful fraction** in sequence.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A motor receives **24 W** and operates at **75%** efficiency. It lifts a **3 kg** load at steady speed with g=**10**. **Find the speed** and wasted power.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3